Year 11 Foundation

REVISION PACK

Algebra 8

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Algebra 8 · Simultaneous equations (Lesson 12.1)

Q	Working	Answer
multiplying both equations		
1.1	$3x + 2y = 14$ $2x + 3y = 11$ $\times 3:\ 9x + 6y = 42$ $\times 2:\ 4x + 6y = 22$ $\text{subtract}:\ 5x = 20$ $x = 4$ $3 \times 4 + 2y = 14$ $y = 1$	$x = 4, y = 1$
1.2	$5x + 2y = 9$ $3x + 4y = 11$ $\times 4:\ 20x + 8y = 36$ $\times 2:\ 6x + 8y = 22$ $\text{subtract}:\ 14x = 14$ $x = 1$ $5 \times 1 + 2y = 9$ $y = 2$	$x = 1, y = 2$
1.3	$2x + 5y = 11$ $3x + 2y = 11$ $\times 2:\ 4x + 10y = 22$ $\times 5:\ 15x + 10y = 55$ $\text{subtract}:\ 11x = 33$ $x = 3$ $2 \times 3 + 5y = 11$ $y = 1$	$x = 3, y = 1$

1.4	$4x + 3y = 18$ $3x + 2y = 13$ $\times 2: \ 8x + 6y = 36$ $\times 3: \ 9x + 6y = 39$ $\text{subtract}: \ x = 3$ $4 \times 3 + 3y = 18$ $y = 2$	$x = 3, y = 2$
1.5	$2x + 3y = 14$ $5x + 4y = 21$ $\times 4: \ 8x + 12y = 56$ $\times 3: \ 15x + 12y = 63$ $\text{subtract}: \ 7x = 7$ $x = 1$ $2 \times 1 + 3y = 14$ $y = 4$	$x = 1, y = 4$
1.6	$3x + 4y = 10$ $2x + 5y = 9$ $\times 5: \ 15x + 20y = 50$ $\times 4: \ 8x + 20y = 36$ $\text{subtract}: \ 7x = 14$ $x = 2$ $3 \times 2 + 4y = 10$ $y = 1$	$x = 2, y = 1$
different signs		
1.7	$3x + 2y = 9$ $4x - y = 1$ $\times 2: \ 8x - 2y = 2$ $\text{add}: \ 11x = 11$ $x = 1$ $3 \times 1 + 2y = 9$ $y = 3$	$x = 1, y = 3$

1.8	$5x + 3y = 7$ $2x - y = 5$ $\times 3: \ 6x - 3y = 15$ $\text{add}: \ 11x = 22$ $x = 2$ $5 \times 2 + 3y = 7$ $y = -1$	$x = 2, y = -1$
1.9	$2x + 3y = 9$ $5x - 2y = 13$ $\times 2: \ 4x + 6y = 18$ $\times 3: \ 15x - 6y = 39$ $\text{add}: \ 19x = 57$ $x = 3$ $2 \times 3 + 3y = 9$ $y = 1$	$x = 3, y = 1$
1.10	$3x + 2y = 8$ $5x - 4y = 6$ $\times 4: \ 12x + 8y = 32$ $\times 2: \ 10x - 8y = 12$ $\text{add}: \ 22x = 44$ $x = 2$ $3 \times 2 + 2y = 8$ $y = 1$	$x = 2, y = 1$
1.11	$4x + y = 6$ $3x - 2y = -1$ $\times 2: \ 8x + 2y = 12$ $\text{add}: \ 11x = 11$ $x = 1$ $4 \times 1 + y = 6$ $y = 2$	$x = 1, y = 2$

1.12	$2x + 5y = 3$ $3x - y = 13$ $\times 5:\ 15x - 5y = 65$ $\text{add}:\ 17x = 68$ $x = 4$ $2 \times 4 + 5y = 3$ $y = -1$	$x = 4, y = -1$
word problems		
1.13	$2w + 3t = 3$ $3w + t = 2.75$ $\times 3:\ 9w + 3t = 8.25$ $\text{subtract}:\ 7w = 5.25$ $w = 0.75$ $2 \times 0.75 + 3t = 3$ $t = 0.5$	**wrap: 0.750 KD** **tea: 0.500 KD**
1.14	$4p + n = 2.25$ $2p + 3n = 1.75$ $\times 3:\ 12p + 3n = 6.75$ $\text{subtract}:\ 10p = 5$ $p = 0.5$ $4 \times 0.5 + n = 2.25$ $n = 0.25$	**pen: 0.500 KD** **notebook: 0.250 KD**
1.15	$3s + 2c = 6$ $2s + 5c = 6.75$ $\times 5:\ 15s + 10c = 30$ $\times 2:\ 4s + 10c = 13.5$ $\text{subtract}:\ 11s = 16.5$ $s = 1.5$ $3 \times 1.5 + 2c = 6$ $c = 0.75$	**sandwich: 1.500 KD** **coffee: 0.750 KD**

1.16	$2c + w = 3$ $c + 3w = 2.75$ $\times 3:\ 6c + 3w = 9$ $\text{subtract}:\ 5c = 6.25$ $c = 1.25$ $2 \times 1.25 + w = 3$ $w = 0.5$	**cake: 1.250 KD** **water: 0.500 KD**
1.17	$3a + 4o = 3$ $2a + o = 1.5$ $\times 4:\ 8a + 4o = 6$ $\text{subtract}:\ 5a = 3$ $a = 0.6$ $3 \times 0.6 + 4o = 3$ $o = 0.3$	**apple: 0.600 KD** **orange: 0.300 KD**
1.18	$5p + 2e = 3.4$ $3p + 4e = 4$ $\times 4:\ 20p + 8e = 13.6$ $\times 2:\ 6p + 8e = 8$ $\text{subtract}:\ 14p = 5.6$ $p = 0.4$ $5 \times 0.4 + 2e = 3.4$ $e = 0.7$	**pencil: 0.400 KD** **eraser: 0.700 KD**